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Increasing Data Availability for Solid Waste Collection Using an IoT Platform based on LoRaWAN and Blockchain

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Abstract

Solid waste management is one of the problems that has gained greater prominence in recent years. This topic has been discussed with greater emphasis in conjunction with environmental issues, driven by changes in the most diverse fields provided by the recent post-pandemic scenario. However, due to traditional or inefficient waste management approaches, most trash bins placed in cities can be seen as overflowing. Therefore, a remote monitoring system is needed to alert the level of garbage in bins to the relevant authority for waste clearance. In this scenario, the Internet of Things (IoT) paradigm plays a vital role in improving smart city applications by tracking and managing city processes. This paper implements a solution based on Low-Power Wide-Area Network (LPWAN) and blockchain technologies to provide the required data available for increasing efficiency in solid waste collection. Finally, this paper provides an evaluation of the proposed architecture related to latency and throughput metrics.

Keywords: Solid waste management; IoT; LoRaWAN; Blockchain

1. Introduction

The management of solid waste is one of the problems that have been gaining greater prominence in recent years, discussed with greater emphasis in conjunction with sustainable development objectives outlined by the document "The 2030 Agenda for Sustainable Development", adopted by all United Nations Member States in 2015 [1]. This document discusses actions under development or developed by countries driven by changes in more diverse fields provided by the recent post-pandemic scenario. A massive part of the solid waste produced is not properly discarded or even returned to the production process from recycling and the use of technologies that increase the effectiveness

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of solid waste management in large urban centers is of paramount importance as confirmed by ”The International Resource Panel” launched by the ”United Nations Environment Programme” [2]. This premise constitutes a strong challenge for the competent authorities that must act holistically, integrating multiple technological solutions that contribute to the active management of information.

The use of technologies developed based on Internet of Things (IoT) becomes essential for the gradual transformation of large urban centers around the world into smart cities [3] [4]. If waste is not properly disposed off, it may lead to huge health issues and it may have adverse effects on the environment also. As a consequence of the growing urban population, waste generated by human is also increasing and millions of tons of waste are being produced by people all over the world every day. Among all, waste collection and transportation are one of the costliest stages in solid waste management [5], as the truck driver must go to each bin every single day and check whether the bin is full or not. If the bin is not full, it is not only a waste of time but also a waste of fuel used by the truck. It increases pollution due to smoke released from trucks and it is not economic because more men are needed for checking all the bins on different routes.

To face the problem described above, we are proposing the use of sensors to measure the amount of waste in the bin, and LoRaWAN technology to send the data for further information processing. When the trash inside the bin reaches a certain threshold level, its location and the current position of the truck driver are detected. Both pieces of information are sent to be processed and the shortest path between them is shown. Using this approach, the garbage bins can be emptied before the dustbin overflow.

Despite the proposal to use smart bins is not new, our research is focused on providing a high degree of data availability of the information sent by the sensors. However, the current specification of LoRaWAN points to some security vulnerabilities as, a centralized authentication process, which under a denial-of-service (DoS) attack, for example, all services may become unavailable. This work proposes a distributed authentication process, based on blockchain, to provide a highly available IoT platform.

2. Related Work

Smart cities bring the need to use goods, services, and natural resources in an optimal way. In this sense, smart monitoring is an important step in achieving this goal [6] and the Internet of Things (IoT) is an important tool for reaching the connection of devices present in these complex systems to improve the efficiency to predict failures, reduce downtime and increase the automation of decision-making processes [8]. Among the processes used to define intelligent monitoring are computational methods to monitor, control, manage and optimize the network of devices with relevant information [7]. With the advancement of communication technology, IoT networks are considered the basis for the functioning of city services.

Also, given the importance of the information that transits between sensors, network servers, and application servers, Blockchain technology comes as an interesting alternative to guarantee the security of the system. This is due to its ability to work with systems endowed with heterogeneity, even though there are concerns about the consumption of power, storage space of the blocks, and hardware utilization, many works have been written [9].

As previously said, waste management constitutes a complex problem nowadays [10]. According to the fact that large urban areas in the world will have an increasing effort the managers to make the use of natural and economic resources more optimized based on smarter systems, as seen in recent papers [11] [12] [13]. However, unlike previous works in the literature, this research proposes a highly available system for the communication of sensors over long distances (LPWAN) used in the dumps, based on blockchain technology to ensure communication in a safer and more scalable way for use by the competent authorities and companies.

3. Overview of LPWAN and Blockchain

3.1. Low Power Area Network

Low Power Wide Area Network (LPWAN) is one of the enabling technologies of the Internet of Things (IoT). The LPWAN is a network designed to allow long-range communications among smart devices, at low bit-rates [14]. This type of network applies to IoT scenarios where low-cost devices need to transmit small messages over long distances (a few kilometers) and transmission delay is tolerable by the application. In LPWANs, end devices are connected to a gateway node directly, thus forming a star topology. The simplified design of LPWANs and duty cycling provide a significant reduction in power consumption. Moreover, the great majority of communications in an LPWAN occur from the end node to the gateway (uplink). This enables end-devices to enter sleep mode after sending a message and remain sleeping for long time intervals. Currently, several LPWAN solutions are available, among them, LoRa [15] stands out as one of the most promising LPWAN technologies.

LoRa is a wireless modulation technique, that provides low-power and long-range communication for small low-cost devices. The LoRa stack can be divided in two layers: physical (PHY) and media access control (MAC). The LoRa PHY layer is a proprietary spread spectrum modulation technique designed by Semtech and it is based on a variation of the Chirp Spread Spectrum (CSS) modulation [16].

LoRaWAN is an open MAC layer protocol developed and maintained by the LoRa Alliance [15] and based on a star-of-stars topology. End-devices are connected to the LoRaWAN network through wireless gateways, as presented in Figure 1. The communication between end-devices and gateways is performed using LoRa PHY modulation. Gateways are responsible for relaying uplink messages to the network server. The communication between gateways and the remaining servers is performed over traditional IP networks. The Network Server (NS) is one of the main components of a LoRaWAN network (center of the star topology) and it has many features, such as packet routing from end-devices; frame authentication; data rate adaptation; relaying authentication messages between the end-devices and the join-server; and roaming. The Application Server (AS) handles all the messages sent by the end devices and provides high-level data management to the end-user. The AS may also send downlink messages to the associated end-devices. Finally, the Join Server (JS) is responsible for root keys storage, for Over-the-Air Activation (OTAA) procedure, and derivation of session keys. The JS is a new element in the LoRaWAN network architecture since it was only included in the latest release of the LoRaWAN specification.

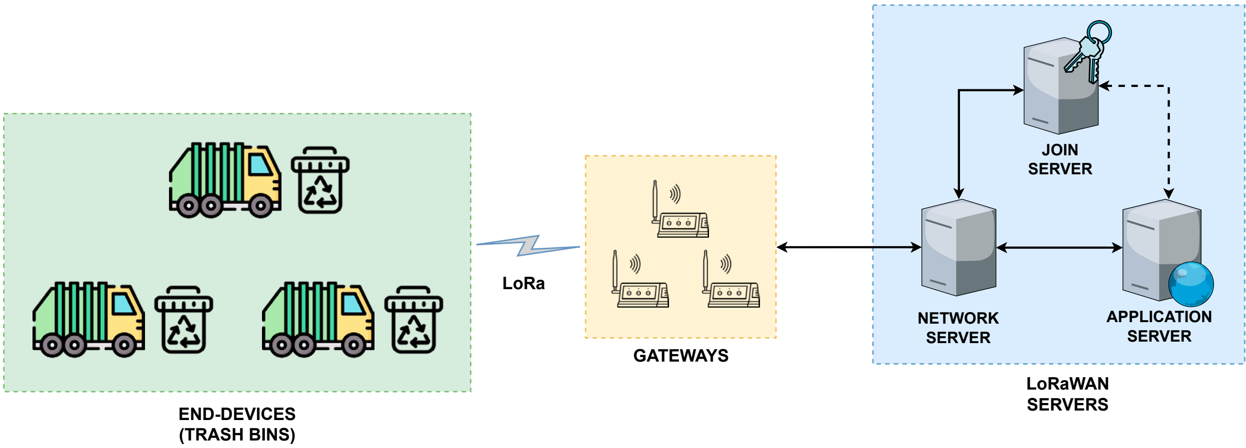

Fig. 1. Illustration of a LoRaWAN Network Architecture.

3.2. Blockchain

Blockchain is a revolutionary technology built on top of a peer-to-peer (P2P) network infrastructure that implements a public distributed ledger where transactions are recorded as an immutable chain of blocks [17]. In other words, a blockchain can be described as a distributed database that stores all digital operations that participating entities execute in a given system. The need for a third party to intermediate a transaction between two participating entities is eliminated with the use of a blockchain due to its decentralized design and the strong cryptographic primitives it is based on. Furthermore, a consensus algorithm is used by the participants to verify each transaction registered in the system. This ensures that the public ledger is tamper-proof. Blockchain also provides audibility for systems, since the information added to the chain can never be altered or erased [17].

There are two types of blockchain: public and permissioned [17]. They basically differ from each other in terms of performance, consensus algorithm, and read permission. In public blockchains, the elevated amount of participating nodes and validations result in decreased throughput and increased latency [18]. Also, any node in the world can make part of the consensus process in public blockchains and all transactions are visible to anyone else. On the other hand, in permissioned blockchains, the smaller number of validations and the limited amount of nodes participating in the consensus process results in much faster transactions. Moreover, only authenticated nodes can participate in the consensus process of permissioned blockchains and only the organization responsible for maintaining the blockchain network can decide whether the transactions are restricted or visible to the public. According to its distributed design, in a blockchain, each participating node stores a copy of the ledger, and blocks of transactions are shared between the nodes.

4. Proposed Architecture for Waste Management

The proposed architecture includes the creation of smart bins, using the Internet of Things (IoT) technology, that gives the current volume and position (latitude and longitude), which will be scattered in several strategic points. Based on this architecture, the system must gather every trash bins coordinates that need to be collected (for the bins which overcame the maximum volume). After that, we can acquire a real-time many-to-many matrix of the elapsed time between the points.

After acquiring the many-to-many matrix for time and identifying the trash bins that need to be collected, we now have sufficient data to define the route. To do that, we model the data using graph theory, where the graph will be created based on the recently obtained many-to-many matrix for time, its vertices, the coordinates of the trash bins, and the edges of the travel time between those coordinates. With the newly created graph, we can now apply a Traveling Salesman Problem (TSP) algorithm to optimize the order of the trash bins to visit. The algorithm receives the following data: the graph newly generated; a routing model responsible for translating every location index, enumerated from 0 to N by default, to the location itself; a distance callback, the purpose is to return the time between any pair of location; and a search parameter, being the heuristic used, by the algorithm, to find certain route. We simulate the situation applied to the city of Fortaleza (Brazil) choosing a scenario with 50 bins to be visited.

However, to promote a highly available IoT Platform the LoRaWAN specification has a serious constraint: the Join Server can be seen as a single point of failure, from a security point of view, as we can see in figure 1. This is due to the centralized design of LoRaWAN and the fact that the Join Server is responsible for handling most of the security tasks, such as the Over-the-Air Activation (OTAA) procedure and storing copies of all encryption keys. A security handshake is performed between an end-device and Join server and Join-Request and Join-Accept messages are exchanged during the procedure and used to derive new session keys. Once OTAA is completed, the new session keys will be used to secure all communications between the end-device and the LoRaWAN servers. The LoRaWAN Backend Interface advises the use of Key Encryption Keys (KEK) and a wrapping algorithm for the secure transportation of session keys between LoRaWAN servers.

Therefore, the Join Server becomes an important target for attackers. If, for example, it suffers a Denial of Service attack, the entire LoRaWAN network can be significantly impacted and stop working. To handle this problem, and increase the availability of the waste management platform, this work proposes to modify the LoRaWAN architecture by replacing the Join Server with a permissioned blockchain infrastructure. In this new architecture, a smart contract is responsible for performing authentication and key management activities as demonstrated in figure 2. The main advantages of our proposal are as follows:

- **Fault-tolerance:** To increase data availability employing a distributed storage provided by multiple peers from a permissioned blockchain, thus preventing the single point of failure issue;
- **Traceability:** To provide system auditability by keeping immutable and verifiable records of all OTAA attempts carried out by end-devices and management operations (e.g., creation and updates of root keys);
- **No need for firmware updates:** No changes are required to the LoRaWAN protocol since all the modifications are done only to the Network Server. All changes occur transparently to the end-devices. Therefore, there is no need for firmware updates to end-devices deployed on a network;
- **No transaction fees:** There are no additional costs in regards to cryptocurrency since the permissioned blockchain is not meant for creating currency.

A working prototype was created as a Proof-of-Concept (PoC) using only open-source tools and comodity hardware. LoPy boards were used as class A end-devices and 1-channel LoRaWAN gateways. Each LoPy board contains an Espressif ESP32 chipset and a Semtech LoRa transceiver SX1276. The LoRaWAN servers were created using the open-source implementations from the ChirpStack project [19]. The permissioned blockchain environment was created with Hyperledger Fabric (v2.0). Golang was chosen as the programming language to implement the chaincode due to its high performance when compared to other languages [20]. Docker images are provided by Hyperledger Fabric to deploy different types of nodes (e.g., peers and orderers). The NS implementation from ChirpStack was modified to invoke the chaincode deployed in the permissioned blockchain.

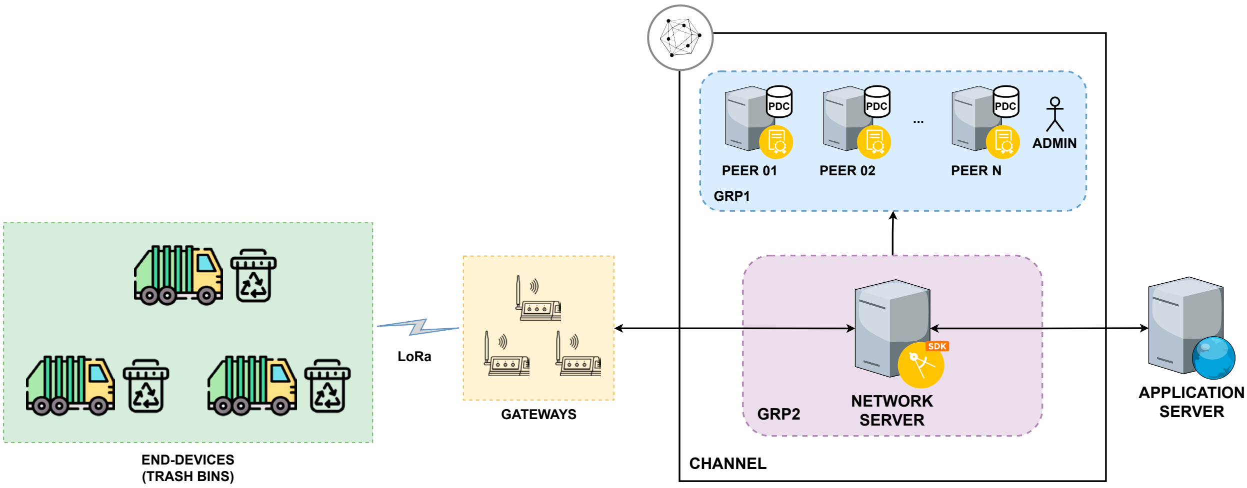

Fig. 2. Proposed LoRaWAN Network Architecture.

5. Performance Analysis

The performance of the working prototype was evaluated in a cloud environment. This section presents the results obtained from performance experiments under different scenarios. Table 1 summarizes the main settings used during the performance analysis. A Golang script was implemented to evaluate the performance of the Hyperledger environment by sending simultaneous requests to the chaincode. Each function is called multiple times and a set of metrics is collected by the performance script. The chaincode was evaluated under different workloads: 10, 50, 100, 250, 500, 750, 1000, 2500, 5000 and 7500 simultaneous requests (for each function). In addition, the performance of the Hyperledger network was evaluated with different numbers of endorsing peers: 2, 4, and 8. The goal was to assess how the number of endorsing peers would affect performance results.

Table 1. Experiments settings

Item	Value
Evaluation metrics	Latency and throughput
Number of requests	10, 50, 100, 250, 500, 750, 1000, 2500, 5000, and 500
Total of executions	15
Confidence interval	95%
Number of peers	2, 4, and 8
Ordering service	Solo
EC2 instances	t2.medium

The working prototype was deployed in an Amazon Virtual Private Cloud (Amazon VPC). Multiple EC2 virtual machines were created to host the Hyperledger Fabric containers (each container was hosted on a separate server) and the LoRaWAN servers. Each virtual machine was configured as a t2.medium EC2 instance (2 vCPUs and 4 GB of RAM) running Ubuntu 18.04. The performance analysis evaluated the following functions:

- **CreateDeviceKeysPrivate:** Stores a new pair of encryption keys in the Private Data Collection (PDC). This function is restricted to the administrator of GRP1 and it should be invoked once a new end-device joins the LoRaWAN network. In order to call this function, a new *Device Keys* object must be passed in the transient field of a transaction.
- **GetDeviceKeysPrivate:** Receives a *DevEUI* value as an input parameter and returns the corresponding *DeviceKeys* object from the PDC. This function is restricted to the administrator of GRP1.
- **HandleJoinRequest:** Receives a serialized Join-Request message as input parameter, derives a new set of session keys, encrypts AppSKey with the pre-shared KEK, and returns the contents of the JoinAns message. This function must be invoked by Network Server during an OTAA procedure.

The following metrics were chosen to evaluate the performance of the proposed architecture:

- **Latency:** For each function call, latency is defined as the amount of time taken by the chaincode to process one request.
- **Throughput:** Defined as the number of requests divided by the total amount of time taken by the chaincode to process all requests in a set.

Figure 3 shows the metric values for the CreateDeviceKeysPrivate function. For up to 2500 requests, the Hyperledger network with 2 endorsing peers has lower latency values than the other two scenarios (Figure 3a). However, a transition begins to occur around 3000 requests. From this point on, a smaller number of endorsing peers starts to suffer impact on its performance while processing a large number of simultaneous requests. After 3000 requests, it is possible to see that scenarios with a larger number of endorsing peers start to show better performance. A second transition occurs around 5000 requests. At this point, Hyperledger networks with 8 endorsing peers start to show better performance than the remaining scenarios. This behavior can be clearly seen through the throughput values (Figure 3b) as the scenario with 8 endorsing peers shows better values than the other scenarios after 5000 requests.

Figure 4 shows the metric values for the GetDeviceKeysPrivate function. Since this function performs only read operations, it shows a different behavior when compared to the other functions. Endorsing peers don't need to exchange messages with each other when performing read-only operations and no new blocks are added to the blockchain, therefore, the chaincode execution is much faster. For up to 1000 requests, all 3 scenarios show similar latency values. However, the values change as the number of requests increases. After 1500 requests, it can be seen that the Hyperledger network with 2 endorsing peers presents a greater degradation when compared to the other two scenarios. The scenario with 8 endorsing peers shows better values for all the metrics evaluated. Based on the values seen, we may conclude that a large number of endorsing peers will always have a better performance for read-only transactions because each request can be processed separately.

Figure 5 presents the metric values for the HandleJoinRequest function. For up to 1000 requests, the scenario with 2 endorsing peers shows the lowest latency values. However, the performance of 2 endorsing peers degrades quickly from 5000 requests onwards (Figure 5a). After 6000 requests, the scenarios with 4 and 8 peers show lower latency values when compared to 2 peers. For the throughput values (Figure 5b), 2 endorsing peers show better throughput for up to 1000 requests. This behavior changes around 2500 requests when 4 endorsing peers begins to have better throughput values. At 5000 requests, the scenario with 8 peers also shows better results than 2 peers.

We conclude that there is a trade-off between performance and availability when choosing the number of endorsing peers in scenarios with a small number of simultaneous requests. For this type of scenario, a reduced number of endorsing peers shows a better performance due to less messages being exchanged between the peers. However, a small number of peers can be detrimental to data availability. A large set of endorsing peers offers a greater degree of availability at the expense of longer execution times. This situation is reversed in scenarios with a large number of requests in which the performance of a small set of peers degrades. Therefore, using multiple endorsing peers is best suited for realistic LPWAN scenarios (which are composed of a large number of end-devices and simultaneous transactions).

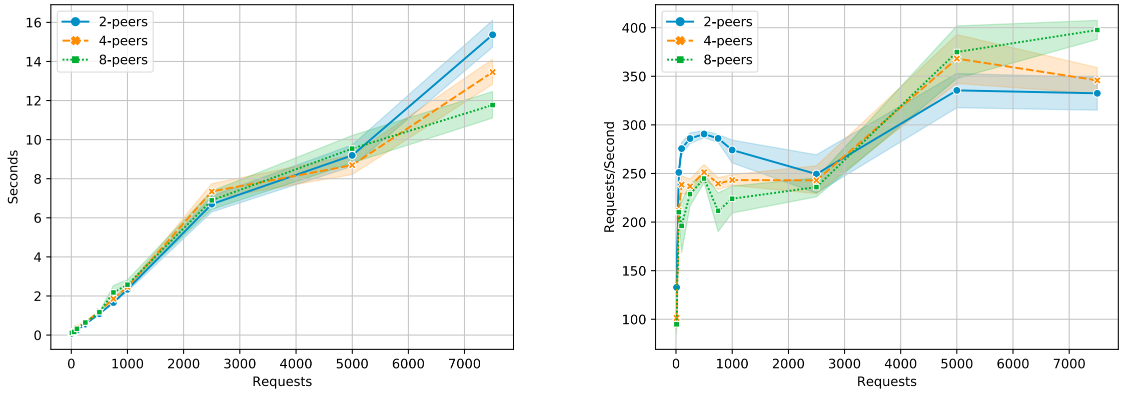

Fig. 3. CreateDeviceKeyPrivate Function (a)Latency (b)Throughput.

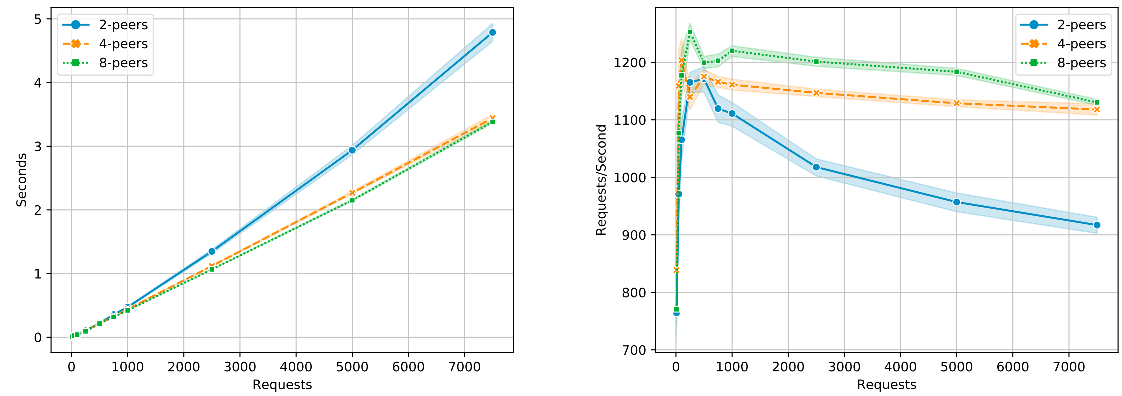

Fig. 4. GetDeviceKeyPrivate Function (a)Latency (b)Throughput.

6. Conclusions

This article presented a smart waste management system by implementing sensors to monitor the amount of trash inside the bin, LoRaWAN communication protocol for low power and long-range data transmission, and blockchain to increase the availability of the architecture by distributing into several nodes the functionalities of key management. The Join Server was removed from the traditional LoRaWAN architecture and replaced by a smart contract. The smart contract is responsible for performing common key management tasks and authentication (OTAA procedure). Every authentication request is stored in the permissioned blockchain and can be verified securely.

Different types of experiments were executed to evaluate the proposed architecture. A performance analysis was performed in a cloud environment with various numbers of endorsing peers and multiple workloads. The results show that there is a trade-off between performance and availability when choosing the number of endorsing peers in small scenarios. A reduced set of endorsing peers demonstrates better throughput values in situations with a few simultaneous requests (up to 2500). However, this behavior is reversed in scenarios with a large number of transactions, as the performance of larger sets of endorsing peers stands out. Therefore, we conclude that using multiple endorsing peers is best suited for realistic LPWAN scenarios.

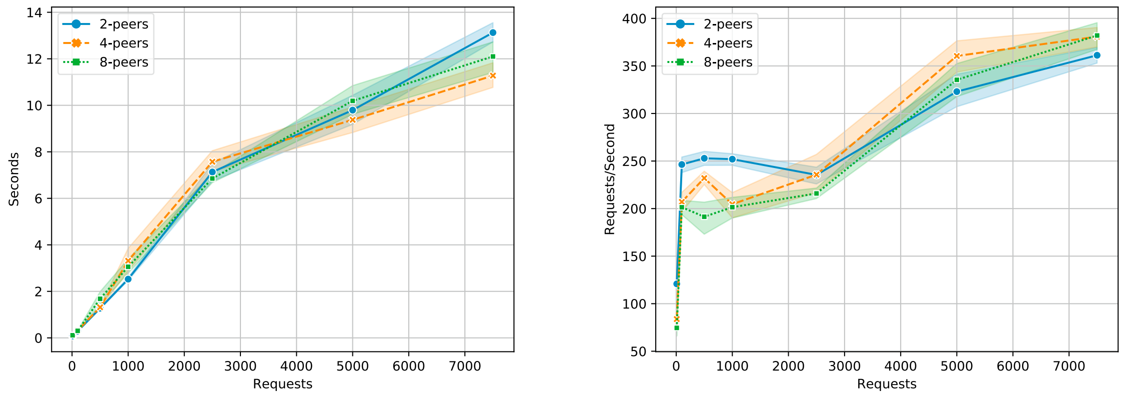

Fig. 5. HandleJoinRequest Function (a)Latency (b)Throughput.

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